



Sequences

A SEQUENCE IS A SERIES OF NUMBERS WRITTEN AS A LIST THAT FOLLOWS A PARTICULAR PATTERN, OR "RULE."

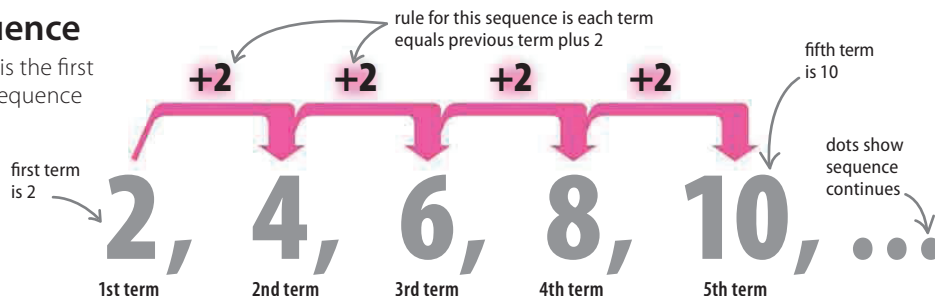
Each number in a sequence is called a "term." The value of any term in a sequence can be worked out by using the rule for that sequence.

The terms of a sequence

The first number in a sequence is the first term, the second number in a sequence is the second term, and so on.

▷ A basic sequence

For this sequence, the rule is that each term is the previous term with 2 added to it.



SEE ALSO

◀ 36–39 Powers and roots

◀ 168–169 What is algebra?

Working with expressions 172–173 ▶

Formulas 177–179 ▶

Finding the "nth" value

The value of a particular term can be found without writing out the entire sequence up until that point by writing the rule as an expression and then using this expression to work out the term.

▷ The rule as an expression

Knowing the expression, which is $2n$ in this example, helps find the value of any term.

$2n$

expression used to find value of term—1 is substituted for n in 1st term, 2 in 2nd term, and so on

means $2 \times n$ substitute 1 for n

$$2n = 2 \times 1 = 2$$

1st term

To find the first term, substitute 1 for n .

$$2n = 2 \times 2 = 4$$

2nd term

To find the second term, substitute 2 for n .

$$2n = 2 \times 41 = 82$$

41st term

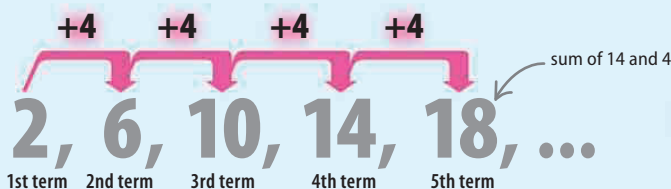
To find the 41st term, substitute 41 for n .

$$2n = 2 \times 1,000 = 2,000$$

1,000th term

For the 1,000th term, substitute 1,000 for n . The term here is 2,000.

In the example below, the expression is $4n - 2$. Knowing this, the rule can be shown to be: each term is equal to the previous term plus 4.



$4n - 2$

expression here is 4 multiplied by n , minus 2

value of term

$$4n - 2 = 4 \times 1 - 2 = 2$$

1st term

To find the first term, substitute 1 for n .

$$4n - 2 = 4 \times 2 - 2 = 6$$

2nd term

To find the second term, substitute 2 for n .

$$4n - 2 = (4 \times 1,000,000) - 2 = 3,999,998$$

1,000,000th term

For the 1,000,000th term, substitute 1,000,000 for n . The term here is 3,999,998.